

LECTURE-18

Monday
13/4/26

Delta Rule: Perceptron learning Algo is for single layer ANN. Delta Rule is for finding new weights of multilayer ANN.

$$W_{\text{new}} = W_{\text{old}} - \eta \frac{dL}{d(W_{\text{old}})}$$

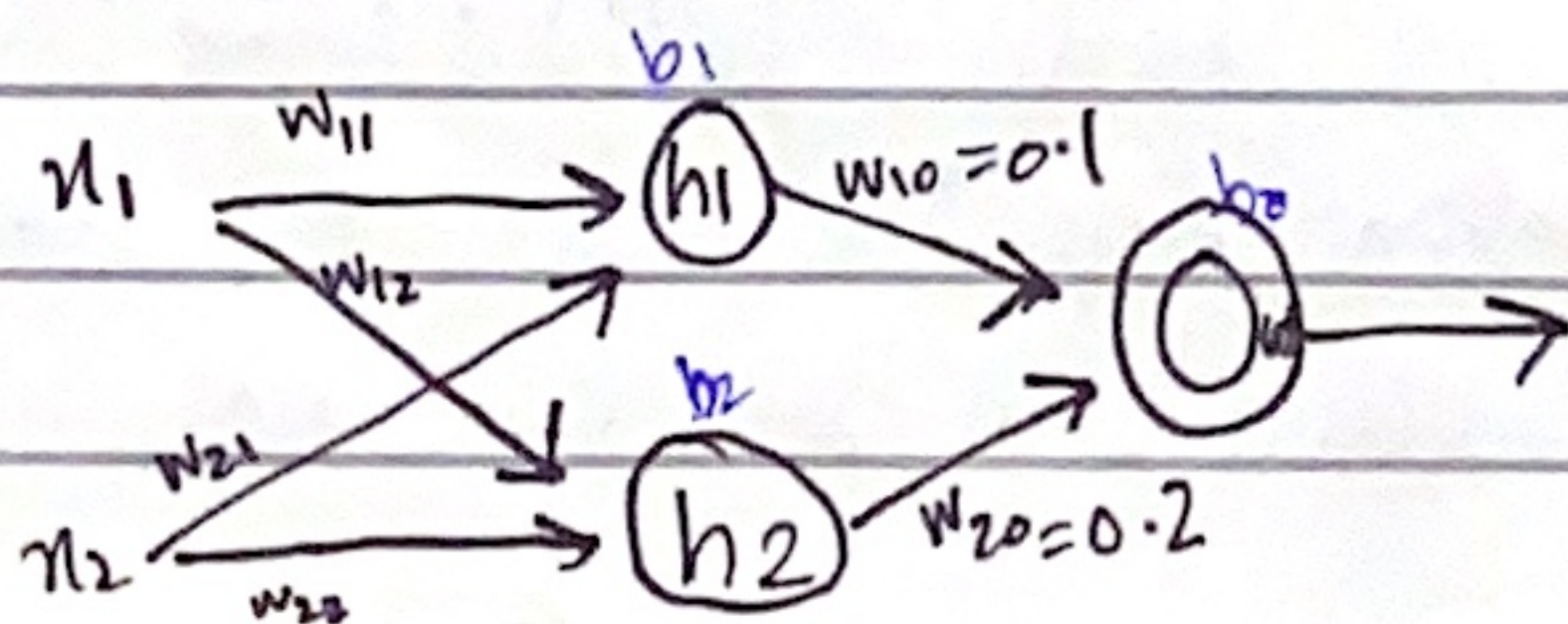
learning rate

$$b_{\text{new}} = b_{\text{old}} - \eta \frac{dL}{d(b_{\text{old}})}$$

$$L = \frac{1}{2} (y - \hat{y})^2$$

$f(y)$ act. funct

★ for getting L in terms of ' w_{10} '; we'll be using chainrule as L is in terms of $\hat{y}$ i.e: $1/(1+e^{-u})$



All weights & biases = 1, $y=1$, $\eta=0.1$

For new weights (w_{10})

$z = \text{Linear Combo.}$

$$(A): \frac{dL}{dw_{10}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{dw_{10}}$$

$$\begin{aligned} \frac{dL}{d\hat{y}} &= \frac{d}{d\hat{y}} \left(\frac{1}{2} (y - \hat{y})^2 \right) \\ &= \frac{1}{2} \cdot 2(y - \hat{y}) \cdot \frac{d}{d\hat{y}} (y - \hat{y}) \\ &= (y - \hat{y}) \cdot (-1) \\ &= \boxed{\hat{y} - y} \end{aligned}$$

$$\begin{aligned} \frac{d\hat{y}}{dz} &= \frac{d}{dz} \left(\frac{1}{1 + e^{-z}} \right) \\ &= \boxed{\hat{y} \cdot (1 - \hat{y})} \end{aligned}$$

act. func. sigmoid
fixed derivative of sigmoid func

* Plug all values in (A)

$$\frac{dL}{dw_{10}} = (\hat{y} - y) \cdot (\hat{y} - (1 - \hat{y})) \cdot h_1$$

$$\begin{aligned} \frac{dz}{dw_{10}} &= \frac{d}{dw_{10}} (w_{10} \cdot h_1 + w_{20} \cdot h_2 + b_0) \\ &= h_1 \cdot 0 + 0 \\ &= \boxed{h_1} \end{aligned}$$

(B): For new bias

$$\begin{aligned} \frac{dL}{db} &= \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz} \cdot \frac{dz}{db} \\ \frac{dz}{db} &= \frac{d}{db} (w_{10} \cdot h_1 + w_{20} \cdot h_2 + b_0) \\ &= \boxed{1} \end{aligned}$$

same

$$(C): \text{For new weights } (w_{11}) : (\hat{y} - y) \cdot (\hat{y} - (1 - \hat{y})) \cdot h_2$$

$h_1 = \text{linear combo} = w_{10} \cdot x_1 + w_{20} \cdot x_2 + b = \boxed{4}$; $h_2 = \text{linear combo} = \boxed{4}$; Sigmoid for h_1 & $h_2 = \boxed{0.982}$;

$O = \text{linear combo} = w_{10} \cdot h_1 + w_{20} \cdot h_2 + b = \boxed{1.2946}$; Sigmoid of $O = \boxed{0.7849} \rightarrow y$

Since $y=1$ but here $y=0.7849$, $\therefore$ find new weights & biases for output (O)

$$(A) w_{10(\text{new})} = 0.1 - 0.1 \left[(0.7849 - 1) \cdot (0.7849 - (1 - 0.7849)) \cdot 0.982 \right]$$

$\rightarrow \eta$

$$w_{10} = \boxed{0.104}$$

$$(B) b_{\text{new}} = 1 - 0.1 \left[(0.7849 - 1) \cdot (0.7849 - (1 - 0.7849)) \right] \cdot 1 = \boxed{1.0036}$$

$$(C) w_{20(\text{new})} = 0.2 - 0.1 \left[(0.7849 - 1) \cdot (0.7849 - (1 - 0.7849)) \right] \cdot 0.982$$

$$w_{20} = \boxed{0.204}$$

Now; we'll change $w_{11}, w_{12}, w_{21}, w_{22}, b_1, b_2$ just like this using h_1 & h_2 . Then we'll finally find new y (which prev. was 0.7849) and continue until answer constant to $y=1$